

**Part A (or B, or C, or MSc)**

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1. Let  $Y \sim p(\cdot|\theta)$  with  $Y = (Y_1, \dots, Y_n)$ ,  $Y \in \mathcal{Y}$  and  $\theta > 0$  a scalar parameter with prior  $\pi(\theta)$ .
- (a) [15 marks] Suppose we observe  $Y = y$  and wish to estimate  $\theta$ .
- (i) Explain what it means for  $\pi(\theta|y)$  to be *doubly intractable* and the consequence for MCMC-based estimation of  $\theta$  given  $Y = y$ .

**Solution:** (B, 4)

$\pi(\theta|y) \propto \pi(y|\theta)\pi(\theta)$ . This is DI if either  $\pi(y|\theta)$  or  $\pi(\theta)$  are intractable functions of  $\theta$ . This may happen if for example the normalisation of  $\pi(y|\theta)$  is not available in closed form. MCMC targeting  $\pi(\theta|y)$  does not require the overall normalisation  $p(y)$  for  $\pi(y|\theta)\pi(\theta)$  as it cancels in the Metropolis ratio  $\pi(y|\theta')\pi(\theta')/\pi(y|\theta)\pi(\theta)$ . However if  $\pi(y|\theta)$  or  $\pi(\theta)$  are intractable then this cannot be evaluated and so MH-MCMC is not possible.

- (ii) Let  $S(y) = (S_1(y), \dots, S_p(y))$  be a vector of summary statistics. For fixed  $\rho > 0$  let  $\Delta_y = \{y' \in \mathcal{Y} : |S(y) - S(y')| \leq \rho\}$ . Let  $\theta \in \Omega$  be discrete. Let  $g(\theta)$  be a probability mass function and let  $M = \max_{\theta} \pi(\theta)/g(\theta)$ . Consider the following ABC algorithm.

Step 1 Simulate  $\theta \sim g(\theta)$ ,  $y' \sim p(y'|\theta)$  and  $u \sim U(0, 1)$ .

Step 2 If  $U \leq \frac{\pi(\theta)}{Mg(\theta)} \mathbb{I}_{y' \in \Delta_y}$  stop and return  $\theta$ , otherwise goto Step 1.

Show that this algorithm targets  $\pi_{ABC}(\theta|y) = \pi(\theta|Y \in \Delta_y)$ .

**Solution:** (S, 7 - slight variant on notes due to introduction of  $g(\cdot)$ )

Let  $p(\theta)$  be the density of a sample returned by this algorithm. Let  $E_y$  be the event  $U \leq \frac{\pi(\theta)}{Mg(\theta)} \mathbb{I}_{y' \in \Delta_y}$ . Now

$$\begin{aligned} P(E_y|\theta) &= \frac{\pi(\theta)}{Mg(\theta)} \int \mathbb{I}_{y' \in \Delta_y} p(y'|\theta) dy', \\ &= \frac{\pi(\theta)}{Mg(\theta)} p(Y \in \Delta_y|\theta). \end{aligned}$$

The marginal probability to reject at step 2,  $r(y)$  say, is

$$r(y) = \sum_{\theta'} g(\theta') (1 - P(E_x|\theta'))$$

so that  $r(y) = 1 - P(\Delta_y)/M$  where  $P(\Delta_y) = \Pr(Y \in \Delta_y)$  is

$$P(\Delta_y) = \sum_{\theta'} \pi(\theta) p(Y \in \Delta_y|\theta).$$

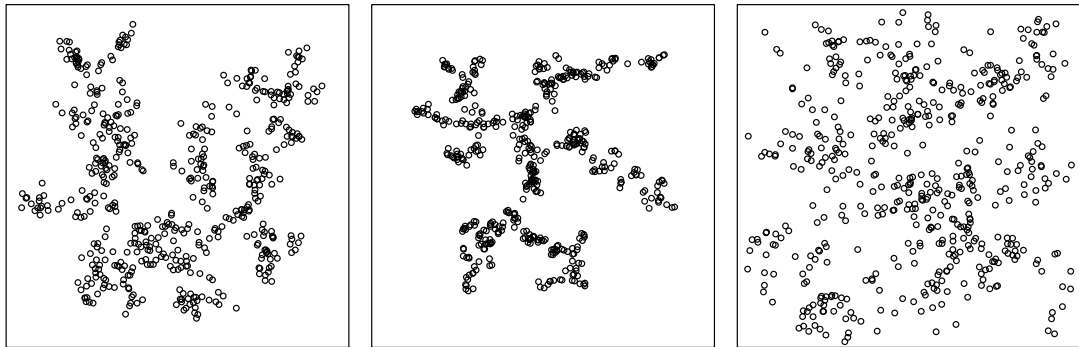
Partitioning on the number of rejections,

$$\begin{aligned} p(\theta) &= g(\theta)P(E_y|\theta) + g(\theta)P(E_y|\theta)r(y) + g(\theta)P(E_y|\theta)r(y)^2 + \dots \\ &= g(\theta)P(E_y|\theta) \times \frac{1}{1 - r(y)} \\ &= \frac{\pi(\theta)p(Y \in \Delta_y|\theta)}{M} \times \frac{M}{P(\Delta_y)} \\ &= \pi(\theta|Y \in \Delta_y). \end{aligned}$$

- (iii) An ABC algorithm yields samples  $\theta^{(t)}, t = 1, 2, \dots, T$  with associated statistics  $S^{(t)} = (S_1^{(t)}, \dots, S_p^{(t)})$ . Explain how to make a regression adjustment to the sampled  $\theta$ -values. What is the purpose of this adjustment?

**Solution:** (B,3) Regress  $\theta^{(t)} \sim S^{(t)}$ . Suppose the estimated regression parameters are  $\hat{\beta}$ . The adjustment is  $\theta_{\text{adj}}^{(t)} = \theta_{\text{adj}}^{(t)} + (S(x) - S^{(t)})\hat{\beta}$ . The adjustment shifts the distribution of  $\theta$  at  $S(X)$  via a linear approximation to the distribution of  $\theta$  at  $S(x)$  which is what we want. The purpose is to improve the approximation, allowing us to use larger bandwidth  $\rho$ .

- (b) [10 marks] Data on locations of large ancient Danish barrow-graves are plotted (left) in the scaled region  $A = [0, 1]^2$ . The following model generates  $n$  grave locations  $y_1, \dots, y_n$ .



For  $i = 1, \dots, n$ , the  $i$ th person dies at a location  $x_i \sim \pi(\cdot|\psi)$  in  $A$ , with  $\psi$  a parameter controlling the distribution of  $x_i$  in  $A$ . The first person to die is buried where they died, at  $y_1 = x_1$ ; the  $i$ 'th person is buried at  $y_i \sim N(y_{j_i}, \sigma^2 I_2)$ , where  $j_i = \arg \min_{k \in 1, \dots, i-1} |x_i - y_k|$  is the nearest pre-existing grave and  $\sigma > 0$  is unknown. Simulations in the figures above show (centre)  $\sigma = 0.02$  and (right)  $\sigma = 0.04$  with  $n = 1000$  with  $\pi(\cdot|\psi)$  uniform.

- (i) Write down the posterior for  $\psi$ ,  $x = (x_1, \dots, x_n)$  and  $\sigma$ , including any missing data needed to make the likelihood well defined.

**Solution:** Consider the missing information. The graves have an arbitrary labeling  $1, \dots, n$  determined by their order in the data vector  $y$ , but we don't know in which order the points/graves were added in the process. This matters because each point except the first is paired with a parent point  $j_i$  which we don't know. Let  $b = (b_1, \dots, b_n)$  give the order, so that for  $i = 1, \dots, n$ , the  $i$ th arrival/grave was individual  $b_i$  who died at  $x_{b_i}$  and is buried at  $y_{b_i}$ , with  $y_{b_1} = x_{b_1}$  for the first arrival. The attachment point for individual  $b_i$  is  $j_{b_i}$  with  $j_{b_i} = \arg \min_{j \in b_1, \dots, b_{i-1}} |x_{b_i} - y_j|$ . In terms of this notation the likelihood is

$$p(y|b, x, \sigma) = \mathbb{I}_{y_{b_1} = x_{b_1}} \prod_{i=2}^n N(x_{b_i} - y_{j_{b_i}}; 0, \sigma^2 I_2),$$

so the posterior is

$$\pi(\psi, b, x, \sigma|y) \propto \pi(\psi, \sigma) \pi(x|\psi) p(y|b, x, \sigma),$$

where I have taken as prior for  $b$  the uniform distribution over permutations of 1 to  $n$ .

- (ii) Assume priors and simulators for all unknown parameters, and for  $y$  given the parameters, are available. Explain how to generate samples  $\psi$  and  $\sigma$  from the ABC posterior. You will need to suggest suitable summary statistics and give an ABC algorithm as pseudo-code.

**Solution:** We simulate synthetic data sets according to the recipe given in the question. Let  $n$  be the number of points in the data.

1. for  $t = 1, \dots, T$

(a) Simulate  $\sigma_{(t)}, \psi^{(t)} \sim \pi(\sigma, \psi)$ .

(b) Simulate  $x_1^{(t)} \sim \pi(x_1 | \psi^{(t)})$  and set  $y_1^{(t)} = x_1^{(t)}$ .

(c) For  $i = 2, \dots, n$

i. Simulate  $x_i^{(t)} \sim \pi(x_i | \psi^{(t)})$  and compute

$$j_i^{(t)} = \arg \min_{k \in 1, \dots, i-1} |x_i^{(t)} - y_k^{(t)}|.$$

ii. Simulate  $y_i^{(t)} \sim N(y_{j_i^{(t)}}^{(t)}, \sigma_{(t)}^2 I_2)$ .

This generates  $T$  sets of parameters  $x^{(t)}, \sigma_{(t)}, \psi^{(t)}$  and associated data vectors  $y^{(t)} = (y_1^{(t)}, \dots, y_n^{(t)})$ , for  $t = 1, \dots, T$ .

Statistics, here are some suggestions.  $S_1(y)$  equals the average over points  $y_1, \dots, y_n$  of the distance to the closest point (should inform  $\sigma$ ).

Another informing  $\sigma$  (the “stickiness” of the outcome) and perhaps  $\psi$ , divide up the window into  $m \times m$  cells. For  $i = 1, \dots, m^2$  let  $V_i$  be the number of graves in cell  $i$  and  $V = (V_1, \dots, V_{m^2})$ . Take as  $S_2$  the variance of  $V$  (use  $m = 10$  for this) or the maximum of  $V$ .

Next take  $m$  small, say  $m = 4$  and take  $S_3 = V_1, \dots, S_{18} = V_{16}$ , just the cell counts. Here  $m$  needs to be small or the dimension of  $S(y) = (S_1(y), \dots, S_p(y))$  will be too large. This should help inform  $\psi$ .

There are many we could consider. Some attention to different statistics informing different parameters. It may be necessary to scale  $S$  so that the different components are on the same scale. Take as distance

$$\delta(s, s') = \sqrt{\sum_{i=1}^p w_i (s_i - s'_i)^2},$$

experimenting with  $w = (w_1, \dots, w_p)$  to get a measure that is sensitive to parameter variation.

The last stage of the ABC analysis is then

2 for  $t = 1, \dots, T$

(a) compute  $\delta^{(t)} = \delta(S(y), S(y^{(t)}))$

3 Set  $\mathcal{T} = \{t \in 1 : T; \delta^{(t)} < \rho\}$ .

4 Output  $\psi^{(t)}, \sigma^{(t)}$  such that  $t \in \mathcal{T}$

Here we choose  $\rho$  so that  $q = T^{-1} \sum_t \mathbb{I}_{\delta^{(t)} < \rho}$  is small (say 0.1 or 0.01). Now the output  $\psi^{(t)}, \sigma^{(t)}$  with  $t \in \mathcal{T}$  are our approximate posterior samples. We check estimates (histograms, posterior means) are insensitive to the choice of  $\rho$ .

2. (a) [8 marks]

(i) Write down the first three of the Savage axioms, in the form given by de Groot.

**Solution:** (B,4) Let  $S$  be a sample space and let  $\mathcal{S}$  be a  $\sigma$ -field of sets in  $S$ . In the following events  $A, B$  etc are subsets of  $S$  and members of  $\mathcal{S}$ .

Axiom 1. For any two events  $A$  and  $B$  exactly one of the following relations must hold:  $A > B$ ,  $A < B$ ,  $A \sim B$ .

Axiom 2. If  $A_1 \cap A_2 = B_1 \cap B_2 = \emptyset$  and  $A_i \geq B_i, i = 1, 2$  then  $A_1 \cup A_2 \geq B_1 \cup B_2$ . If in addition either  $A_1 > B_1$  or  $A_2 > B_2$  then  $A_1 \cup A_2 > B_1 \cup B_2$ .

Axiom 3. If  $A \in \mathcal{S}$  then  $\emptyset \leq A$ .

(ii) For  $y_i \in \{0, 1\}$  for  $i = 1, 2, 3, \dots$ , the sequence of distributions  $\pi_n(y_1, \dots, y_n), n = 1, 2, 3, \dots$  is *marginally consistent* if

$$\pi_n(y_1, \dots, y_n) = \pi_{n+1}(y_1, \dots, y_n, 0) + \pi_{n+1}(y_1, \dots, y_n, 1).$$

Suppose our prior for the events  $Y_1 = y_1, \dots, Y_n = y_n$  is given by  $\pi_n(y_1, \dots, y_n)$  and that these distributions are not marginally consistent. Show that our preferences on the events  $Y_1 = y_1, \dots, Y_n = y_n$  do not satisfy the Savage axioms.

**Solution:** (B, 4, identical to problem sheet) Let  $A_z = \{(Y_1 = y_1, \dots, Y_n = y_n, Y_{n+1} = z)\}$  for  $z \in \{0, 1\}$  for some fixed binary  $y_1, \dots, y_n$ , and let  $A = \{(Y_1 = y_1, \dots, Y_n = y_n)\}$ . If  $\pi_n(y_1, \dots, y_n) \neq \pi_{n+1}(y_1, \dots, y_n, 0) + \pi_{n+1}(y_1, \dots, y_n, 1)$  then we must have either

$$\pi_n(y_1, \dots, y_n) < \pi_{n+1}(y_1, \dots, y_n, 0) + \pi_{n+1}(y_1, \dots, y_n, 1)$$

or

$$\pi_n(y_1, \dots, y_n) > \pi_{n+1}(y_1, \dots, y_n, 0) + \pi_{n+1}(y_1, \dots, y_n, 1).$$

Suppose  $\pi_n(y_1, \dots, y_n) > \pi_{n+1}(y_1, \dots, y_n, 0) + \pi_{n+1}(y_1, \dots, y_n, 1)$ . This corresponds to the preference  $A > A_0 \cup A_1$ . But (as events)  $A$  occurs if and only if  $A_0 \cup A_1$  occurs so  $A \sim A_0 \cup A_1$ . This contradicts the first axiom since exactly one of  $>, <, \sim$  must hold. The other possibility leads to  $A < A_0 \cup A_1$  and  $A \sim A_0 \cup A_1$ .

(b) [9 marks] Consider the following model for a random binary  $n \times n$  matrix  $G$  with diagonal entries all equal zero. For  $i = 1, \dots, n$  let  $z_i = (z_{i,1}, z_{i,2})$  be a point taken independently and uniformly at random in  $[0, 1]^2$ . Let  $z = (z_1, \dots, z_n)$ . The random matrix  $G = G(z)$  is defined as follows. For each  $i, j$  in  $1, \dots, n$  let

$$G_{i,j} = \begin{cases} 1 & \text{if both } z_{i,1} > z_{j,1} \text{ and } z_{i,2} > z_{j,2} \\ 0 & \text{otherwise.} \end{cases}$$

Let  $\Omega_n$  be the set of all  $n \times n$  binary matrices which can be generated in this way and let  $\pi_n(g) = \Pr(G = g)$  for  $g \in \Omega_n$ .

(i) For  $G \in \Omega_n$  let  $G^{(-n)}$  be obtained from  $G$  by deleting the last row and column. Show that if  $G \sim \pi_n$  then  $G^{(-n)} \sim \pi_{n-1}$  and briefly interpret.

**Solution:** (N,3) If  $z^{(-n)}$  is  $z_1, \dots, z_{n-1}$  then  $G(z)^{(-n)} = G(z^{(-n)})$  (the relation between  $j$  and  $k$  is not changed by the presence of  $n$ ). But then for  $g \in \Omega_{n-1}$  we have  $\Pr(G(Z)^{(-n)} = g) = \Pr(G(Z^{(-n)}) = g) = \pi_{n-1}(g)$ . Distributions in the sequence  $\pi_n, n = 1, 2, 3, \dots$  are marginally consistent.

- (ii) Let  $P_n(g) = 1/|\Omega_n|$  for  $g \in \Omega_n$ , so  $P_n$  is the uniform distribution. Show that if  $G \sim P_n$  then  $G^{(-n)}$  is not in general distributed according to  $P_{n-1}$ . *Hint: the set sizes are  $|\Omega_2| = 3$ ,  $|\Omega_3| = 19$  and  $|\Omega_4| = 219$ .*

**Solution:** (N,3) Cant be as 19 is not divisible by 3 (and 219 is not divisible by 19). More formally, the operation  $G^{(-3)}$  of removing the last column of  $G \in \Omega_3$  is a many to one map from elements of  $\Omega_3$  to elements of  $\Omega_2$ . Now  $\Omega_2$  has 3 elements,  $G_0, G_1$  and  $G_2$  say. For  $k = 0, 1, 2$ , let  $n_k = |\{G \in \Omega_3 : G^{(-3)} = G_k\}|$ . Then if  $G \sim P_3$  we have  $\Pr(G^{(-3)} = G_k) = n_k/19$ . But if  $G^{(-3)} \sim P_2$  then  $\Pr(G^{(-3)} = G_k) = 1/3$  and  $19/3 = n_k$  for some integer  $n_k$ , a contradiction, so  $G^{(-3)}$  is not distributed according to  $P_2$ .

- (iii) Suppose that for each  $n = 1, 2, 3, \dots$  our prior for  $n \times n$  binary matrix  $G$  is  $P_n$ . Explain why these preferences over events do not satisfy the Savage axioms.

**Solution:** (N,3) We have seen (in part (a) of this question) that if a family of distributions is not marginally consistent then it will not satisfy the Savage axioms. The proof here is just the same. More generally, or alternatively, if a joint distribution over the binary entries in  $G \sim P_n$  exists then the joint distribution of the entries in  $G^{(-n)}$  is the marginal. Since our prior belief contradicts this (by the result in the previous subpart) the joint distribution cant exist. But a joint distribution exists if and only if the Savage axioms are satisfied, so they cant be satisfied.

- (c) [8 marks] We can think of  $G$  as representing a directed graph with a directed edge from  $i$  to  $j$  if  $G_{i,j} = 1$  and no directed edge from  $i$  to  $j$  if  $G_{i,j} = 0$ . Suppose  $G$  is unknown and our data are  $J$  random walks  $(a_1, \dots, a_K)$  of length  $K$  on the nodes of  $G$ .

1. The first node  $a_1 \sim U\{1, \dots, n\}$  is chosen at random.
2. For  $k = 1, \dots, K - 1$  choose  $a_{k+1} \sim U\{j : G_{j,a_k} \neq 1\}$  at random from nodes  $j$  with no directed edge pointing back from  $j$  to  $a_k$  ( $a_k$  itself is always in this set).

Suppose we observe  $J$  walks  $A_j = (a_{j,1}, \dots, a_{j,K})$ ,  $j = 1, \dots, J$ . Let  $A = \{A_1, \dots, A_J\}$  denote the set of observed walks.

- (i) Write down the posterior  $\pi(z|A)$  in terms of  $G(z)$  and  $A$  for the prior  $G \sim \pi_n$ .

**Solution:** (S, 3)

$$\pi(z|A) \propto \pi(z)L(z; A)$$

with  $\pi(z) = \mathbb{I}_{z \in [0,1]^{2n}}$ . Let  $G = G(z)$  and  $I_{j,k} = \{v : G_{v,a_{j,k}} \neq 1\}$ . We have

$$L(G(z); A) = \prod_{j=1}^J n^{-1} \prod_{k=1}^{K-1} |I_{j,k}|^{-1}.$$

- (ii) Explain how you would use MCMC to sample  $\pi(G|A)$ . Give an explicit algorithm. There is no need to simplify acceptance ratios or check irreducibility.

**Solution:** (S, 3) We can sample  $z^{(t)} \sim \pi(z|A)$ ,  $t = 1, \dots, T$  approximately using MCMC and then set  $G^{(t)} = G(z^{(t)})$  for each  $t = 1, \dots, T$ . A simple scheme for  $z|A$  is to visit each  $z_{i,j}$  in turn for  $i = 1, \dots, n$  and  $j = 1, 2$ . If  $X_t = z$  then we

set  $z' = z$  and then replace  $z_{i,j}$  with a uniform draw  $z'_{i,j} \sim U(0, 1)$  and accept wp

$$\alpha(z'|z) = \min \left\{ 1, \frac{L(G(z'); A)}{L(G(z); A)} \right\}.$$

- (iii) Explain how you would use your MCMC to estimate the posterior probability that all entries in  $G$  are zero, and how you might estimate the unknown true  $G$  itself.

**Solution:** (S, 2) If  $g_0$  is the  $n \times n$  zero matrix, then we estimate  $\pi(g_0|A)$  with  $\hat{\pi}_0 = T^{-1} \sum_t \mathbb{I}_{G(z^{(t)})=g_0}$ . A point estimate for  $G$  is the mode of the output (the average of  $G|A$  is not in general a binary matrix), estimated by  $G^* = \arg \max_G \sum_t \mathbb{I}_{G(z^{(t)})=G}$ .

3. Let  $M > 0$  be a positive integer and let  $\alpha > 0$  be a real positive scalar. Let  $w = (w_1, \dots, w_M)$ . Consider the following model,

$$\begin{aligned} w &\sim \text{Dirichlet}(\alpha_1, \dots, \alpha_M) \quad \text{with } \alpha_m = \alpha/M, m = 1, \dots, M \\ z_i|w &\sim \text{Multinom}(w), \quad i = 1, \dots, n. \end{aligned}$$

Here for  $i = 1, \dots, n$ ,  $z_i \sim \text{Multinom}(w)$  means  $z_i = m, m \in \{1, \dots, M\}$  with probability  $w_m$ .

- (a) [16 marks] Suppose the list  $z_1, \dots, z_n$  of cluster labels contains  $K \leq M$  unique values  $m_1, \dots, m_K$ . For  $k = 1, \dots, K$  let  $C_k = \{i : z_i = m_k, i = 1, \dots, n\}$  and  $C = \{C_1, \dots, C_K\}$ .  
 (i) Show that in this model the probability to generate partition  $C$  is

$$P(C) = \frac{\Gamma(\alpha)}{\Gamma(\alpha/M)^K} \frac{M!}{(M-K)!} \frac{\prod_{k=1}^K \Gamma(\alpha/M + |C_k|)}{\Gamma(\alpha + n)}$$

where  $|C_k|$  is the number of elements in  $C_k$  for  $k = 1, \dots, K$ .

*Hint: The Dirichlet probability density for the probability vector  $w$  is*

$$\text{Dirichlet}(w; \alpha_1, \dots, \alpha_M) = \frac{\Gamma(\sum_{m=1}^M \alpha_m)}{\prod_{m=1}^M \Gamma(\alpha_m)} \prod_{m=1}^M w_m^{\alpha_m - 1}$$

**Solution:** (B, 8, identical to problem sheet)  $C$  is determined by  $z = (z_1, \dots, z_n)$ . However there are many  $z$ 's giving the same  $C$ , as the cluster labels are not important. We need  $K$  distinct labels, in any permutation, so there are  $K! \binom{M}{K}$   $z$ 's for each  $C$ . Given  $w_1, \dots, w_M$   $p(z_i = m|w) = w_{z_i}$ . The marginal probability  $p(z)$  to get a particular set of cluster labels  $z$  is

$$\begin{aligned} p(z) &= E_w \left( \prod_{i=1}^n w_{z_i} \right) \\ &= \int \frac{\Gamma(\sum_m \alpha_m)}{\prod_m \Gamma(\alpha_m)} \prod_{m=1}^M w_m^{n_m + \alpha_m - 1} dw_1 \dots dw_M \\ &= \frac{\Gamma(\alpha)}{\Gamma(\alpha/M)^M} \frac{\prod_{m=1}^M \Gamma(n_m + \alpha/M)}{\Gamma(n + \alpha)} \end{aligned}$$

where  $n_m = \sum_i \mathbb{I}_{z_i=m}$  counts the number of  $z$ 's equal  $m$  and  $\alpha_m = \alpha/M$  so  $\sum_m n_m + \alpha_M = n + \alpha$  etc. We want to sum this over all  $z$ 's that determine a given partition  $C = (C_1, \dots, C_K)$ . Let  $n'_k = |C_k|$ . These  $z$ 's must match the cluster sizes in  $C$  and the number of clusters  $K$ , so  $M - K$  of the  $n_m$ 's are zero and the rest must match the  $|C_k|$  values, so

$$\prod_{m=1}^M \Gamma(n_m + \alpha/M) = \Gamma(\alpha/M)^{M-K} \prod_{k=1}^K \Gamma(|C_k| + \alpha/M).$$

Then

$$p(z) = \frac{\Gamma(\alpha)}{\Gamma(\alpha/M)^K} \frac{\prod_k \Gamma(|C_k| + \alpha/M)}{\Gamma(n + \alpha)}.$$

This is the same for all  $z$  that determine  $C$  (ie, it doesn't depend on the specific cluster labels  $m \in \{1, 2, \dots, M\}$ ), so

$$P(C) = K! \binom{M}{K} \frac{\Gamma(\alpha)}{\Gamma(\alpha/M)^K} \frac{\prod_k \Gamma(|C_k| + \alpha/M)}{\Gamma(n + \alpha)},$$

as supposed.

- (ii) Let  $C_{-i} = \{C_{-i,1}, \dots, C_{-i,K_{-i}}\}$  be the partition we obtain by removing  $i$  from its set in  $C$ . The partition  $C_{-i}$  contains  $K_{-i}$  sets  $C_{-i,k}, k = 1, \dots, K_{-i}$  following the removal of  $i$ . Here  $K_{-i}$  may be less than  $K$  since if  $i$  is in a set by itself we remove that set entirely.

Suppose we observe  $C_{-i}$  and we consider the distribution of  $C|C_{-i}$ . Show that

$$P(i \in C_k | C_{-i}) \propto \begin{cases} \frac{\alpha}{M} + |C_{-i,k}| & \text{if } k \leq K_{-i} \text{ and} \\ \frac{\alpha}{M}(M - K_{-i}) & \text{if } k = K_{-i} + 1. \end{cases}$$

*Hint: the Gamma function satisfies  $x\Gamma(x) = \Gamma(x+1)$ .*

**Solution:** (N, 5) We saw that

$$P(C) \propto \frac{1}{\Gamma(\alpha/M)^K} \frac{1}{(M-K)!} \prod_{k=1}^K \Gamma(\alpha/M + |C_k|)$$

Consider  $P(C)/P(C_{-i})$ . If  $i$  is in a cluster by itself then  $K = K_{-i} + 1$  and there is an extra factor in the product with  $|C_K| = 1$  so

$$P(C)/P(C_{-i}) \propto \frac{1}{\Gamma(\alpha/M)} \frac{1}{(M-K)} \Gamma(\alpha/M + 1)$$

which gives the second line using  $x\Gamma(x) = \Gamma(x+1)$ . If  $i$  is in cluster  $C_k$  then  $K = K_{-i}$  then  $|C_k| = |C_{-i,k}| + 1$  in one of the terms in the product and nothing else changes so

$$P(C)/P(C_{-i}) \propto \Gamma(\alpha/M + |C_k|)/\Gamma(\alpha/M + |C_{-i,k}|),$$

and this is  $\alpha/M + |C_{-i,k}|$  using the same identity.

- (iii) Find a generalisation of the Chinese restaurant process generating random partitions  $C$  of  $n$  elements according to  $P(C)$ .

**Solution:** (N,3) Customer 1 is seated at table 1. Now suppose after the arrival of  $j-1$  customers that  $K$  tables are occupied and for  $k = 1, \dots, K$  that  $n_k$  customers are seated at table  $k$ . Customer  $j$  sits at table  $k \in \{1, \dots, K\}$  with probability proportional to  $n_k + \alpha/M$ , and sits at a new table, table  $K+1$ , with probability proportional to  $\alpha(1 - K/M)$ .

- (b) [9 marks] Now consider a mixture model for data  $y = (y_1, \dots, y_n)$ . Suppose  $C \sim P(C)$  is a random partition of  $1, \dots, n$  into  $K$  sets. The cluster to which  $y_i$  belongs is  $z_i = \{k : i \in C_k\}$ . Each cluster has an observation model parameter  $\theta_k^*$  distributed as

$$\theta_k^* | C \sim H, \quad k = 1, \dots, K$$

independent and identically distributed according to  $H$ , where  $H$  is a distribution on  $\mathbb{R}$  with probability density  $h$  say. Let  $\theta^* = (\theta_1^*, \dots, \theta_K^*)$ . The observation model is

$$y_i | C, \theta^* \sim f(y_i | \theta_{z_i}^*), \quad i = 1, \dots, n,$$

independent given  $C$  and  $\theta^*$ , with  $f$  the density of a real scalar variable and  $y \in \mathbb{R}^n$ .

- (i) Write down the posterior  $\pi(C, \theta^* | y)$  for  $C, \theta^*$  given  $y$ .

**Solution:** (S,3)

$$\pi(C, \theta^* | y) \propto P(C) \prod_{k=1}^K h(\theta_k^*) \prod_{i \in C_k} f(y_i | \theta_k^*)$$

Let  $\mathcal{C}_K$  be the set of partitions of  $1, \dots, n$  with exactly  $K$ . The parameter space  $(C, \theta^*) \in \Omega$  is

$$\Omega = \bigcup_{K=1}^{\min(n, M)} \bigcup_{C \in \mathcal{C}_K} \mathbb{R}^K \times \{C\},$$

and the normalising constant is

$$p(y) = \sum_{K=1}^{\min(n, M)} \sum_{C \in \mathcal{C}_K} \int_{\mathbb{R}^K} P(C) \prod_{k=1}^K h(\theta_k^*) \prod_{i \in C_k} f(y_i | \theta_k^*) d\theta_1^* \dots d\theta_K^*.$$

- (ii) Give a Gibbs sampler updating  $C, \theta^*$  to  $D, \psi^*$  which takes each  $i = 1, \dots, n$  in turn and assigns it a new cluster by sampling the conditional distribution for the new cluster assignment  $P(i \in D_k | C_{-i}, \theta_{-i}^*, y)$ , explaining how  $\psi^*$  is determined.

**Solution:** (N,6) For each  $i = 1, \dots, n$  remove  $i$  from the set containing it and assign it to cluster  $k$  according to the conditional distribution

$$P(i \in C_k | C_{-i}, y) \propto \begin{cases} \frac{\alpha}{M} + |C_{-i, k}| f(y_i | \theta_k^*) & \text{if } k \leq K_{-i} \text{ and} \\ \frac{\alpha}{M} (M - K_{-i}) f(y_i | \theta_{K_{-i}+1}^*) & \text{if } k = K_{-i} + 1, \end{cases}$$

with  $\theta_{K_{-i}+1}^* \sim H$  an independent draw from  $H$  and  $\psi^* = (\theta^*, \theta_{K_{-i}+1}^*)$  if we draw  $k = K_{-i} + 1$ , and  $\psi^* = \theta^*$  if  $1 \leq k \leq K_{-i}$ .